Experiment 9 : Experiment to understand Matplotlib library use cases in Data Science

through visualization

Aim :

Experiment to understand Matplotlib library use cases in Data Science

through visualization

Description: Use Iris data set to understand the Matplot lib library

Program Code :

```
from google.colab import files

uploaded = files.upload()
import pandas as pd
import matplotlib.pyplot as plt
import seaborn as sns

%matplotlib inline

# Read dataset
data = pd.read_csv('/content/Iris_Dataset.csv')

data

# Dataset information
data.info()

# Statistical description
data.describe()

# Count varieties
```

```
data.value_counts('variety')

# Count plot
sns.countplot(x='variety', data=data)
plt.show()

# Convert categorical variable into dummy variables
dummies = pd.get_dummies(data.variety)

FinalDataset = pd.concat(
    [pd.get_dummies(data.variety),
     data.iloc[:, [0, 1, 2, 3]]],
    axis=1
)

FinalDataset.head()

# Scatter plot - Sepal Length vs Sepal Width
sns.scatterplot(
    x='SepalLengthCm',
    y='SepalWidthCm',
    hue='variety',
    data=data
)

plt.xlabel('SepalLengthCm')
plt.ylabel('SepalWidthCm')
plt.show()
```

```
# Scatter plot - Petal Length vs Petal Width
sns.scatterplot(
    x='PetalLengthCm',
    y='PetalWidthCm',
    hue='variety',
    data=data
)

plt.xlabel('PetalLengthCm')
plt.ylabel('PetalWidthCm')
plt.show()

# Pair plot
sns.pairplot(data, hue='variety', height=3)
plt.show()

# Histogram - Petal Length
sns.FacetGrid(
    data,
    hue='variety',
    height=5
).map(
    sns.histplot,
    'PetalLengthCm'
).add_legend()

plt.show()

# Histogram - Petal Width
```

```
sns.FacetGrid(
    data,
    hue='variety',
    height=5
).map(
    sns.histplot,
    'PetalWidthCm'
).add_legend()

plt.show()

# Histogram - Sepal Length
sns.FacetGrid(
    data,
    hue='variety',
    height=5+++
).map(
    sns.histplot,
    'SepalLengthCm'
).add_legend()

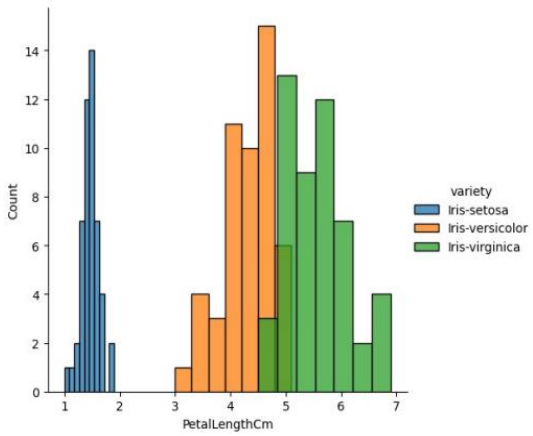
plt.show()

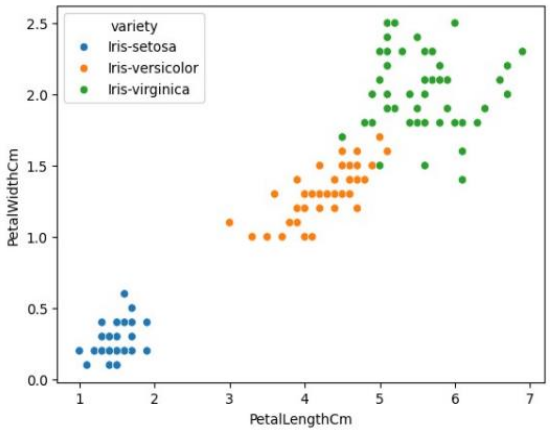
# Histogram - Sepal Width
sns.FacetGrid(
    data,
    hue='variety',
    height=5
).map(
```

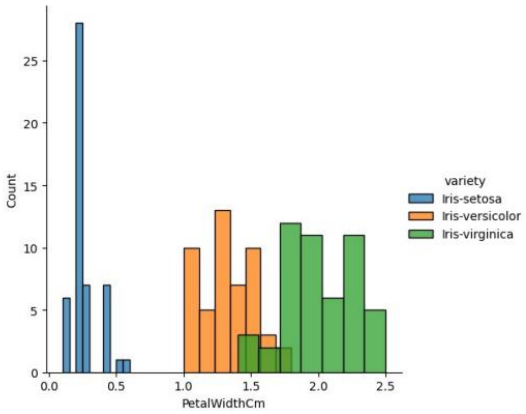
```
    sns.histplot,
    'SepalWidthCm'
).add_legend()

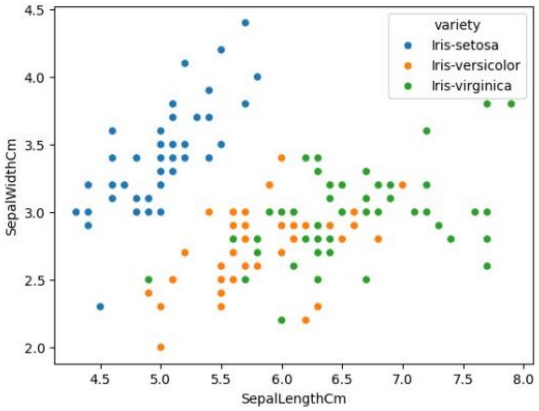
plt.show()
```

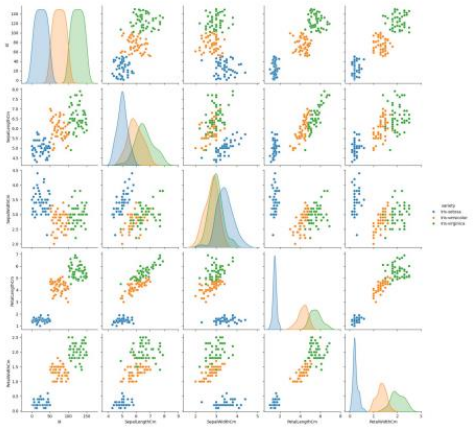
Output :

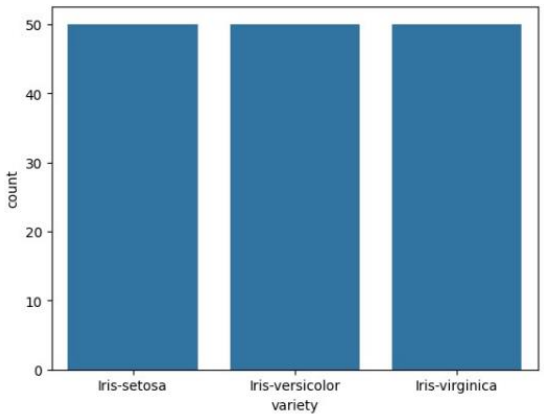


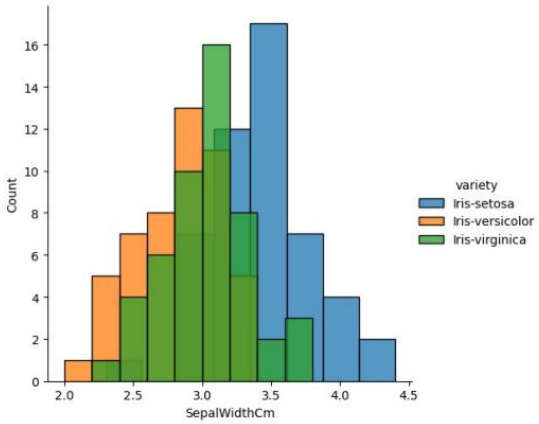


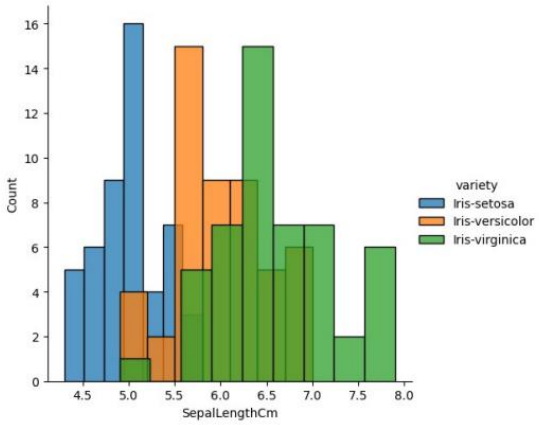












Result :

Thus,we Experiment to understand Matplotlib library use cases in Data Science through visualization